Solutions to linear systems

1 It is the point of intersection of two lines.

2 By substituting the ordered pair into each equation and seeing if it satisfies them.

3 Yes it is a solution.

4 Yes it is a solution.

5 No it is not a solution.

6 No it is not a solution.

7

a No **b** Yes

8

a Yes **b** No **c** No

9

a Yes **b** No

10

a Yes **b** No **c** No

11

a No **b** No **c** Yes

Table of values method

12 $x = 7, y = 14$

13 $x = -6, y = 12$

14 $x = 4, y = -1$

15 $x = -5, y = -15$

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16 $x = 4, y = 5$

17

a

x	-1	0	1	2	3
y	-3	0	3	6	9

b

x	-1	0	1	2	3
y	17	15	13	11	9

c $x = 3, y = 9$

18

a

x	-7	-6	-5	-4	-3
y	28	24	20	16	12

b

x	-7	-6	-5	-4	-3
y	31	26	21	16	11

c $x = -4, y = 16$

19

a $x + y = 5$

x	1	2	3
y	4	3	2

$y = 2x - 1$

x	1	2	3
y	1	3	5

b $x = 2, y = 3$

20

a

x	1	3	5
y	5	15	25

b

x	1	3	5
y	5	-3	-11

c $x = 1, y = 5$

21

a

x	$y = 3x - 1$	$y = -5x + 23$
1	2	18
2	5	13
3	8	8
4	11	3
5	14	-2

b $x = 3, y = 8$

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22

a

x	$y = -2x - 8$	$y = 5x + 34$
-7	6	-1
-6	4	4
-5	2	9
-4	0	14

b $x = -6, y = 4$

Applications

23

a 4GB **b** $74 **c** BZ&Z

24

a

Number of classes attended in one year	Cost for the Annual Membership ($)	Cost to Drop in for every class($)
0	840	0
10	840	210
20	840	420
30	840	630
40	840	840
50	840	1050

b 40

c Annual membership

d 'Drop in' to every class

25

a

Hours of work	Amount charged by plumber A	Amount charged by plumber B
4	160	160
5	177	195
6	194	230
7	211	265

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b 4 hours

26

a

Number of bowls made in one month	Cost to make (dollars)	Revenue from sales (dollars)
0	400	0
4	500	200
8	600	400
12	700	600
16	800	800
20	900	1000

b 16

c 800

d Profit

e $75

27 $x + y = 11500, y = 4x$

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